

Melt Grafting of Geometry-Tailored Voltage Stabilizers for High-Performance Polypropylene Insulation

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High-performance polypropylene (PP) insulation is essential for next-generation high-voltage direct current (HVDC) systems. This study presents a scalable, solvent-free melt grafting strategy to covalently incorporate thermally stable aromatic voltage stabilizers (VSs)—2-vinylnaphthalene (VN), 1,1-diphenylethylene (DPE), and 4-vinylbiphenyl—into isotactic PP. VSs are rationally selected based on their high boiling points, π -conjugated aromatic structures, and vinyl functionality, enabling compatibility with melt processing and uniform bulk functionalization of PP. Electrical tests showed that VN and DPE markedly improved volume resistivity, suppressed leakage current, and enhanced DC breakdown strength, while their effects diminished at loadings above 1.0 wt.%, respectively. Thermally stimulated depolarization current confirmed that these gains originated from deeper and well-distributed trap formation. Quantum chemical and finite element simulations further revealed how geometry-tailored VSs modulate trap formation and suppress space charge accumulation, linking molecular-scale trapping to macroscopic charge dynamics. This work highlights a practical and recyclable route to engineer PP insulation with superior insulation reliability, offering new insights into stabilizer structure-property relationships for HVDC applications.

1. Introduction

The increasing deployment of energy storage systems, power electronics, and devices operating under extreme environmental conditions has created a growing demand for high-performance electrical insulation materials.^[1,2] Among these, polymer-based dielectrics are recognized for their superior electrical properties, including high dielectric strength, low dielectric loss, and excellent voltage endurance.^[3] Their lightweight nature, cost-effectiveness, and ease of processing make them ideal for applications such as high-voltage direct current (HVDC) cables, capacitors, and embedded insulation in power modules.^[4]

Polymeric materials are also gaining prominence due to their recyclability, particularly when compared to traditional ceramic insulators and thermosetting resins. Thermoset polymers, such as epoxy, polyimide, and crosslinked polyethylene (XLPE), have been widely used in

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high-voltage components—including transformers, bushings, and cable insulation—due to their outstanding thermal and dielectric properties.^[5–7] However, their permanently crosslinked structure hinders reprocessability and recyclability, posing a challenge in light of recent efforts toward carbon neutrality and circular material systems.^[8,9] As a result, thermoplastic materials ranging from commodity plastics to engineering plastics have received increasing attention.

Polypropylene (PP) has gained widespread attention as an insulating material in various applications, including power cables, capacitors, and electrical components.^[10–12] It offers a unique combination of low dielectric loss, high volume resistivity, high breakdown strength (BDS), good thermal properties, and chemical resistance.^[13–15] Its recyclability further enhances its appeal over thermoset-based systems, such as XLPE. These characteristics position PP as a promising candidate for next-generation eco-friendly HV insulation systems.

Despite these advantages, PP's intrinsic limitations remain a concern, particularly under high-voltage DC stress. Its non-polar molecular structure lacks deep trap sites necessary to mitigate space charge accumulation.^[15,16] This deficiency leads to field distortion, charge injection, and reduced dielectric reliability over prolonged use. Consequently, several strategies have been proposed to enhance the electrical performance of PP, particularly in suppressing space charge and improving BDS.

Beyond its electrical shortcomings, PP also suffers from mechanical rigidity. Homo PP exhibits high stiffness and low toughness,^[17,18] making it unsuitable for cable insulation applications that require flexibility and softness. To address this issue, researchers have developed elastomer-toughened PP blends using materials such as ethylene-propylene rubber, polyolefin elastomer, styrene-ethylene-butadiene-styrene, and ethylene-propylene-diene monomer.^[19–22] These blends reduce modulus and improve flexibility, depending on the elastomer content. However, elastomeric inclusions have been reported to negatively affect insulation performance due to their relatively high electrical conductivity and reduced BDS.^[23,24]

Blends of different polyolefins, such as low-density PE, high-density PE, and PP, have also been investigated to impart mechanical flexibility while contributing to improved dielectric performance, although their effectiveness remains somewhat limited in achieving simultaneous optimization of both properties.^[25,26]

To further enhance electrical reliability, the incorporation of organic voltage stabilizers (VSs) has gained increasing attention due to their chemical compatibility and effectiveness in enhancing electrical performance.^[27,28] Organic molecules containing delocalized π -electrons can capture high-energy electrons, thereby reducing electrical degradation.^[29] A wide range of π -conjugated structures, including styrene, tetracene, and pentacene, has been explored as small-molecule stabilizers.^[30] Furthermore, conjugated polymers such as polystyrene and polythiophene have also shown promise in suppressing space charge and improving dielectric reliability.^[31–33]

However, small-molecule VSs often exhibit limited stability under high-voltage and high-temperature conditions. They are prone to volatilization, migration, and poor dispersion in the polymer matrix, leading to long-term instability. To mitigate these drawbacks, researchers have explored the covalent graft-

ing of VSs directly onto polymer backbones to anchor them in place and ensure uniform distribution. For instance, Yuan et al. grafted styrene onto isotactic PP (iPP), resulting in improved DC BDS and volume resistivity.^[34] Similarly, Liang et al. reported that PP grafted with 4-acetoxystyrene exhibited enhanced space charge suppression and electrical insulation performance.^[35] Despite their effectiveness, most of these grafting methods require solution-based chemical synthesis, which often entails complex processes and the use of organic solvents. As a result, they hinder scalability and make practical implementation challenging.

In this work, we introduce a one-pot melt grafting strategy employing thermally stable aromatic VSs with high boiling points. Specifically, 2-vinylnaphthalene (VN), 1,1-diphenylethylene (DPE), and 4-vinylbiphenyl (VBP) were selected due to their benzene ring-based aromatic structures, which facilitate effective charge trapping, as well as their vinyl functional groups and high boiling points, making them suitable for melt mixing with iPP.

To evaluate the impact of grafted VSs on insulation performance, we systematically investigated key electrical properties, including volume resistivity, leakage current, and DC BDS at room temperature. In addition, thermally stimulated depolarization current (TSDC) measurements were conducted to assess trap characteristics, while quantum chemical simulations provided insights into trap depth and the electronic structure changes induced by each stabilizer. Furthermore, time-resolved charge transport was analyzed using the finite element method (FEM) simulation to link molecular-level trapping behavior with macroscopic charge dynamics.

Overall, this study presents a practical, scalable, and environmentally conscious approach to enhancing the dielectric performance of PP by the covalent incorporation of high-performance aromatic voltage stabilizers. The findings offer valuable insights into the structure–property relationships of voltage stabilizers and pave the way for the development of next-generation thermoplastic insulating materials for HVDC insulation applications.

2. Results and Discussion

2.1. Concept

Conventionally, VS grafting onto iPP involves solvent-based chemical reactions, as illustrated on the left side of **Figure 1**. In this multi-step process, iPP is first oxidized in a liquid-phase reaction to introduce free radicals, which then serve as active sites for stabilizer attachment. While this approach enables chemical bonding, it suffers from several drawbacks: it is time-consuming, environmentally unfriendly due to solvent use, and often results in surface-selective modification rather than uniform incorporation. As a result, the stabilizer is poorly distributed, leading to inconsistent performance—particularly in large-scale applications.

To overcome these limitations, this study employed a solvent-free, one-step melt grafting method (**Figure 1**, right). In this approach, iPP was directly blended with dicumyl peroxide (DCP) as a radical initiator and selected VS molecules in a high-shear internal mixer at elevated temperatures. The thermal decomposition of DCP generates free radicals on the iPP backbone, which subsequently react with the vinyl groups of the stabilizers to form covalent bonds. As a result, the VSs were stably grafted onto the

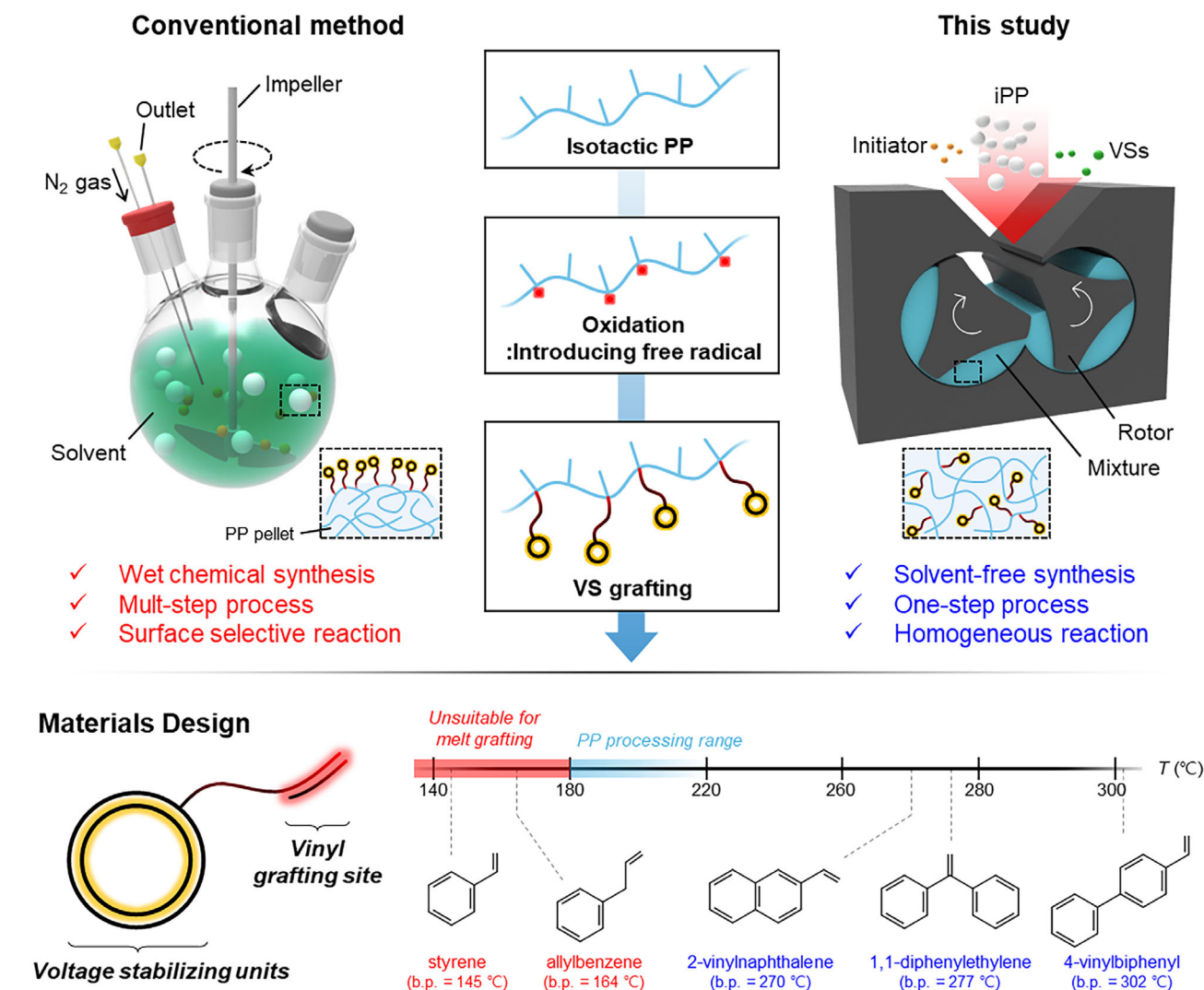


Figure 1. Schematic illustration comparing traditional solution-phase VS grafting with the proposed one-step melt grafting method for iPP. The bottom panel shows the boiling points of selected VS candidates in relation to the PP processing temperature window.

iPP chains with consistently high yield (Table S1, Supporting Information). This melt-phase process enables bulk grafting under industrially relevant conditions, eliminating the need for solvents or pretreatment, and ensuring homogeneous dispersion of the stabilizer throughout the polymer matrix.

A key novelty in this study lies in the rational design and selection of stabilizer candidates. As illustrated in the lower part of Figure 1, the boiling points of potential stabilizers are critically important. PP is typically processed in the range of 180–250 °C. Therefore, only stabilizers with boiling points above this range are suitable for melt grafting, as they remain in the polymer melt phase throughout the process. Compounds such as styrene (b.p. 145 °C) and allylbenzene (b.p. 164 °C), which have been previously reported as VSs, are unsuitable due to their lower boiling points and higher vapor pressures (Table 1), which may result in significant loss through evaporation during processing and pose potential safety hazards such as flammability or volatilization-induced pressure build-up. In contrast, VN (b.p. ≈270 °C), DPE

(b.p. ≈277 °C), and VBP (b.p. ≈302 °C) exhibit high thermal stability, low vapor pressure, and remain reactive during melt mixing. Furthermore, the incorporation of two benzene rings in their structures is anticipated to facilitate the formation of deep trap sites, thereby enhancing their effectiveness as VSs.

This design strategy ensures that the selected VSs not only withstand the melt-grafting process but also function effectively within the polymer matrix. By aligning thermal stability, molecular reactivity, and dielectric functionality, this study presents a practical and scalable route for engineering high-performance iPP-based insulation materials for HVDC applications.

2.2. Conduction Behavior

The volume resistivity of VS-grafted PP is presented in Figure 2a and Table 2. Pristine iPP exhibited a volume resistivity of

Table 1. Physical properties of selected VS candidates.

	Styrene	Allylbenzene	VN	DPE	VBP
Boiling point [°C (at 760 mmHg)]	145.2 ± 7.0	157.3 ± 10.0	270.9 ± 7.0	277.2 ± 10.0	301.7 ± 12.0
Vapour pressure [mmHg (at 25 °C)]	6.2 ± 0.1	3.6 ± 0.1	0.0 ± 0.3	0.0 ± 0.3	0.0 ± 0.3
Vaporization enthalpy [kJ mol ⁻¹]	36.6 ± 0.8	37.8 ± 0.8	48.9 ± 0.8	49.5 ± 0.8	52.0 ± 0.8

$3.53 \times 10^{17} \Omega \text{ cm}$ at 25 °C under an electric field of 10 kV mm^{-1} for 1800 s. Among the tested VSs, VN consistently exhibited the highest resistivity across all contents, indicating superior charge-blocking performance owing to effective trap formation and charge immobilization. DPE also showed relatively high resistivity, particularly at 0.3 and 0.7 wt.%, despite some variation with content. In contrast, VBP displayed a marked decrease beyond 0.3 wt.%, approaching the level of neat iPP at higher concentrations. Overall, the slight decline in resistivity at higher grafting levels may be attributed to excessive grafting, which can introduce structural defects or conductive pathways that reduce the insulating performance.^[36]

Figure 2b further supports this observation by presenting the leakage current behavior of the same samples under DC electric fields of 5, 10, and 15 kV mm^{-1} . At 5 kV mm^{-1} , neat iPP

exhibits the lowest leakage current, indicating strong inherent insulation under mild electric stress. However, under a field of 10 kV mm^{-1} , VN-grafted PP demonstrates the most effective suppression of leakage current, likely due to the formation of deeper and more uniform trap sites within the polymer matrix. DPE also shows improved performance in this field, while VBP and neat iPP record higher current levels. At 15 kV mm^{-1} , the leakage current increases across all samples, with pristine iPP again exhibiting the lowest current—possibly due to its simpler structure, which avoids conduction pathways introduced by grafted additives. These shifts in performance confirm that the dielectric behavior of each formulation is not constant but is strongly influenced by the applied field, underscoring the importance of evaluating insulation materials under a range of operating conditions.

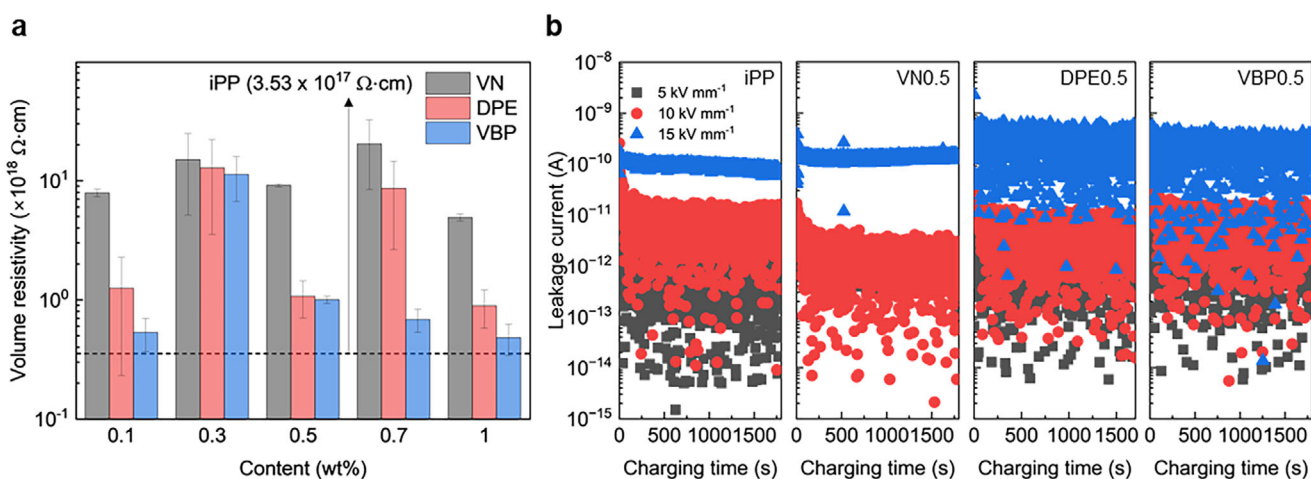


Figure 2. a) Volume resistivity of VS-grafted PP samples, calculated from the leakage current measured at 25 °C under an electric field of 10 kV mm^{-1} for 1800 s. b) leakage current of VS-grafted PP samples as a function of electric field. The dark gray squares, red circles, and blue triangles represent the data measured at 5, 10, and 15 kV mm^{-1} , respectively.

Table 2. Volume resistivity of VS-grafted PP samples.

VS content [wt.%]	VN	DPE	VBP
	Volume resistivity [$\times 10^{18} \Omega \text{ cm}$]	Volume resistivity [$\times 10^{18} \Omega \text{ cm}$]	Volume resistivity [$\times 10^{18} \Omega \text{ cm}$]
0	0.35	0.35	0.35
0.1	7.88	1.25	0.53
0.3	15.00	12.80	11.30
0.5	9.12	1.07	1.00
0.7	20.40	8.61	0.68
1.0	4.90	0.89	0.48

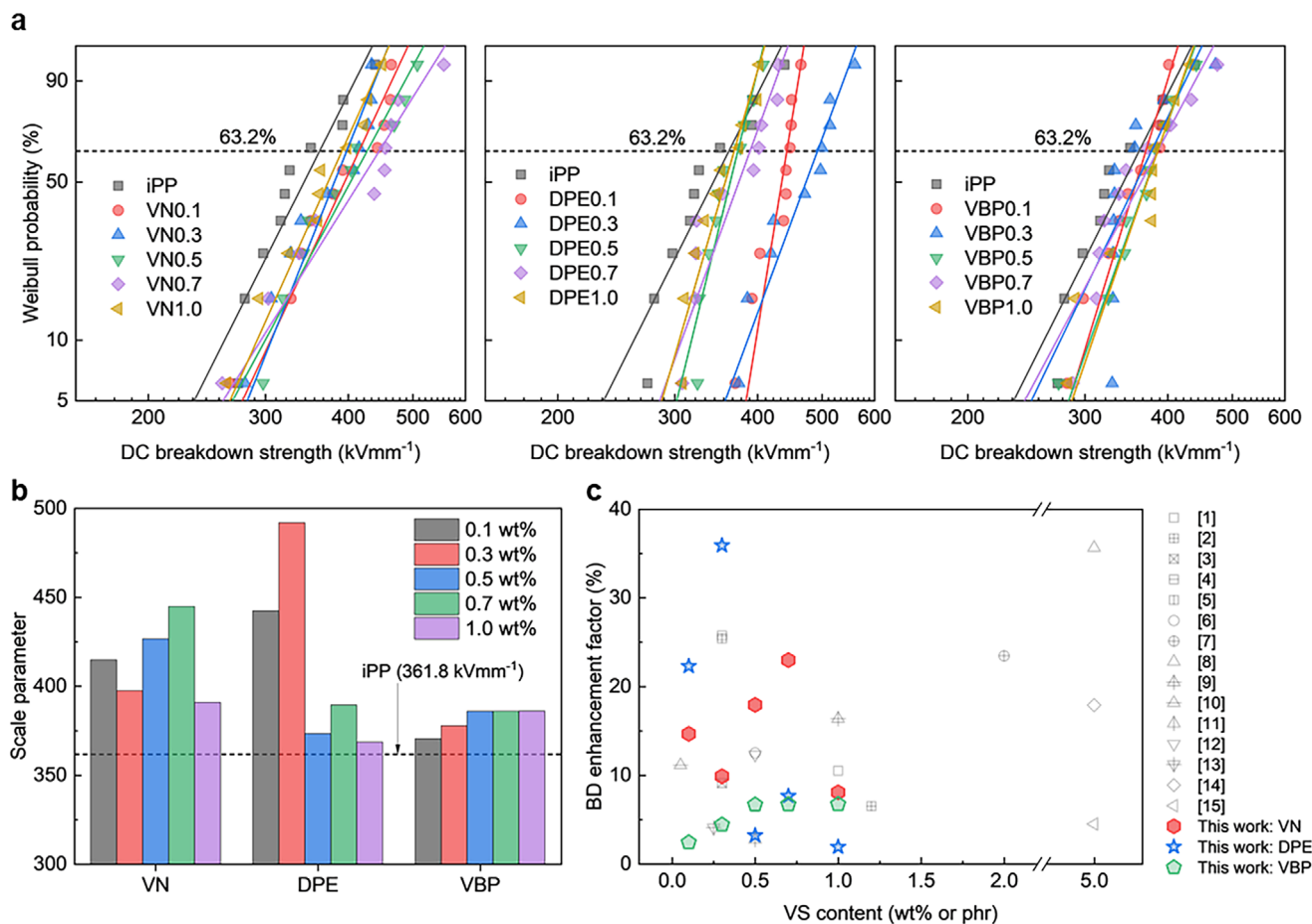


Figure 3. a) Weibull probability plots of DC BDS for iPP and VS-grafted PP samples. The dotted line indicates the characteristic BDS, corresponding to 63.2% cumulative probability. b) Scale parameter of VS-grafted PP samples. c) Comparison of DC BDS enhancement factors for various polymer insulators grafted with VSs, including newly developed melt-grafted VS samples.

These findings highlight that VN outperforms VBP and DPE in improving the volume resistivity of iPP; however, the extent of improvement depends on both the applied electrical field and the optimal grafting contents. Beyond the optimal level, excessive grafting may introduce undesirable conductive pathways, thereby reducing the overall insulation performance. Therefore, careful selection and tuning of the voltage stabilizer type and content are essential to balance trap formation, charge suppression, and long-term dielectric performance under high-voltage DC stress.

2.3. DC Breakdown Behavior

Figure 3 and **Table 3** present the DC BDS characteristics of iPP grafted with VN, DPE, and VBP at loadings of 0.1, 0.3, 0.5, 0.7, and 1.0 wt.%. The BD behavior was evaluated using the Weibull statistical method, where the scale parameter represents the characteristic BDS at 63.2% probability of failure, which serves as a key indicator of BDS under DC electric field stress.

For VN-grafted PP, incorporation of 0.1 wt.% VN increased the BDS compared to pristine iPP. The highest performance

was observed at 0.7 wt.%, with a peak scale parameter of 444.90 kV mm⁻¹. At 1.0 wt.%, the value slightly declined to 390.92 kV mm⁻¹, indicating a saturation point beyond which additional VN may reduce insulation performance.

DPE-grafted PP samples exhibited a sharp increase in BDS at 0.3 wt.%, reaching a maximum scale parameter of 491.97 kV mm⁻¹—the highest among all tested samples. However, further increases in DPE content resulted in performance degradation, with BDS values declining to 373.29 kV mm⁻¹ at 0.5 wt.%, 389.41 kV mm⁻¹ at 0.7 wt.% and 368.59 kV mm⁻¹ at 1.0 wt.%. This trend suggests that DPE is highly effective in enhancing BDS at low contents but becomes less beneficial at higher loadings.

Meanwhile, the VBP-grafted PP samples showed modest increases in BDS across all contents. At 1.0 wt.%, the scale parameter increased to 386.03 kV mm⁻¹, a slight improvement over the 361.84 kV mm⁻¹ of neat iPP.

Figure 3b presents a direct comparison of the scale parameters for iPP samples grafted with different VSs across various grafting contents. DPE at 0.3 wt.% yields the highest BDS among all samples, followed by VN at 0.7 wt.%, which also shows a significant improvement. In contrast, VBP exhibits only moderate

Table 3. DC breakdown scale parameter and shape parameter of VS-grafted PP samples.

VS content [wt.%]	VN		DPE		VBP	
	Scale parameter [kV mm ⁻¹]	Shape parameter	Scale parameter [kV mm ⁻¹]	Shape parameter	Scale parameter [kV mm ⁻¹]	Shape parameter
0	361.84	6.39	361.84	6.39	361.84	6.39
0.1	414.80	7.36	442.40	21.04	370.47	11.19
0.3	397.49	8.63	491.97	9.29	377.73	7.14
0.5	426.63	6.38	373.29	14.02	385.88	9.63
0.7	444.90	5.49	389.41	9.49	385.96	6.45
1.0	390.92	7.67	368.59	11.83	386.03	9.99

variation with increasing content, resulting in limited enhancement in BDS. The nonlinear trends observed for all VSs suggest that excessive additive loading can diminish performance due to over-saturation, poor dispersion, or residual unreacted components.

This tendency is consistent with the morphological observations (Figure S1, Supporting Information). VN-g-PP showed uniform dispersion up to 0.5 wt.% but local aggregation at 1.0 wt.%, while DPE-g-PP exhibited only a few very small dispersed features at 0.1 wt.% but pronounced irregular aggregation at 1.0 wt.%, indicating the formation of VS-rich regions due to limited compatibility. In contrast, VBP-g-PP maintained a uniform morphology without visible aggregation. In addition, a slight red shift in the UV-vis spectra further supports these trends—arising from aggregation-induced π - π interactions in VN and DPE, and from electronic coupling in VBP (Figure S2, Supporting Information).

Figure 3c further compares the DC breakdown performance of the VS-grafted PP samples with various stabilizers reported in the literature (the detailed values of BDS and BD enhancement factor are presented in Table S2 in Supporting Information). Among all tested stabilizers, DPE at 0.3 wt.% delivers the highest BDS, highlighting its exceptional efficiency at low contents. Apart from the DPE-grafted sample, no other stabilizer exhibits a comparable BDS enhancement at such low content. VN shows its highest performance at 0.7 wt.%, reflecting a broader

optimal range than DPE. VBP, while not significantly increasing BDS, shows consistent behavior across concentrations, reinforcing its role in stabilizing dielectric properties rather than maximizing strength. The benchmarking results underscore that melt-grafted stabilizers—particularly DPE and VN—can match or exceed the performance of solution-grafted stabilizers at lower loadings, demonstrating the practical effectiveness of the melt-grafting approach in developing high-performance iPP-based insulation materials for HVDC applications.

2.4. Trap Level Distribution

To better understand the role of aromatic stabilizers in charge trapping behavior, TSDC measurements were performed, and the trap level distributions were calculated from the resulting data. The trap density vs trap depth plots for pristine iPP and VS-grafted PP samples are shown in Figure 4. Pristine iPP exhibits a baseline peak at 1.17 eV, corresponding to intrinsic deep traps associated with crystalline-amorphous interfaces and molecular disorder within the polymer matrix.

For VN-grafted PP (VN-g-PP), the trap depth increases to 1.21 eV at 0.1 wt.%, suggesting the formation of deeper traps that can more effectively capture and localize charge carriers. This is consistent with the behavior of π -conjugated aromatic rings, which are known to enhance charge localization through

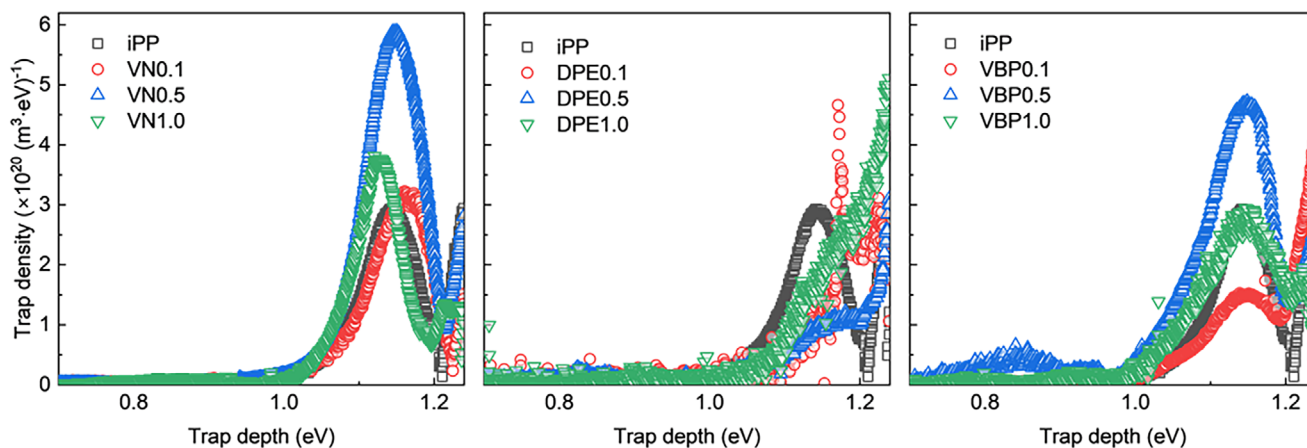


Figure 4. TSDC plots of iPP and VS-grafted PP samples.

electronic polarization.^[37] At a higher loading of 1.0 wt.%, however, the trap depth decreases to 1.12 eV, indicating the development of shallower and less uniform traps. Such reduction at elevated content is commonly associated with limited dispersion or structural perturbations, which restrict effective trap formation.^[38] At the intermediate loading of 0.5 wt.%, VN exhibits a deep trap depth of 1.147 eV together with the highest trap density ($\approx 5.85 \times 10^{20} \text{ (m}^3 \text{ eV)}^{-1}$) among the stabilizers at the same loading, suggesting that the deep trap depth combined with a large population of effective sites underpins the strong charge-localization observed for VN ≈ 0.5 wt.%.

For DPE-g-PP, the trap depth increases monotonically with content, shifting from 1.20 eV (0.1 wt.%) to 1.23 eV (1.0 wt.%), while the 0.5 wt.% point (1.167 eV) appears as a local deviation from strict monotonicity with relatively low deep trap density ($1.08 \times 10^{20} \text{ (m}^3 \text{ eV)}^{-1}$) compared with other molecules. This deepening suggests progressive trap formation, likely aided by the extended aromatic structure of DPE, which can promote stronger charge localization. However, the relatively low trap density implies limited availability of effective trapping sites, and at higher contents, less uniform grafting or molecular dispersion may also contribute to heterogeneous trap distribution.

For VBP-g-PP, the trap distribution remains largely similar to that of pristine iPP, with only slight variations. At 0.1 wt.%, the main peak is located ≈ 1.15 – 1.23 eV, closely overlapping with iPP, indicating that VBP grafting introduces little change in trap depth. At 0.5 wt.%, VBP exhibits a deep-trap level of 1.149 eV together with a moderate trap density ($4.66 \times 10^{20} \text{ (m}^3 \text{ eV)}^{-1}$), suggesting only a modest modification of the intrinsic trap landscape. When the content is increased to 1.0 wt.%, the peak shifts marginally to 1.14 eV, reflecting a minor reduction in trap depth and reaffirming that the overall VBP effect on deep traps is limited.

The modification of trap characteristics by VSs in the iPP matrix is closely linked to the observed improvements in electrical performance. At a given loading, the electrical response reflects a balance between the deep-trap level and the population of effective trapping sites (density). For VN, the combination of an adequate deep trap depth and the highest trap density at 0.5 wt.% yields robust carrier immobilization and suppresses long-range transport. This behavior accounts for the generally parabolic trend—exhibiting low leakage current and superior BDS within the intermediate loading range (0.1–1.0 wt.%) (Figures 2 and 3).

In contrast, although DPE develops the deepest traps with increasing content, its electrical performance degrades sharply at higher loadings. This degradation is likely associated with the pronounced phase separation and heterogeneous dispersion observed in scanning electron microscopy (SEM) images, suggesting that morphological factors, rather than intrinsic trap characteristics, dominate its insulating behavior at elevated concentrations (Figure S1, Supporting Information).^[39]

2.5. Quantum Chemical Simulation

To complement the experimental findings and provide molecular-level insights into the trap and insulation behavior of VS-grafted PP samples, quantum chemical simulations

Table 4. Energy parameters for iPP, VN, DPE and VBP.

Energy parameter	iPP	VN	DPE	VBP
Electron affinity, χ (eV)	−1.05	1.14	0.69	0.82
Ionization energy, ϕ_i (eV)	7.65	5.88	6.51	6.38
Energy gap, E_g (eV)	8.67	4.74	5.82	5.56

were conducted to analyze their electronic structure and charge localization characteristics. We performed density functional theory (DFT) calculations at the B3LYP/6-311G(d,p) level with D3BJ dispersion to analyze the electronic structure and charge localization in iPP and the VN/DPE/VBP-grafted systems.

Table 4 summarizes the electronic structure properties of iPP and its derivatives grafted with various aromatic groups (VN, DPE, and VBP), including the electron affinity (χ), ionization energy (ϕ_i), and energy gap (E_g). The energy gap of iPP was calculated to be 8.67 eV, which is comparable to the experimentally reported bandgap of bulk PP (≈ 8.2 eV)^[40] and slightly lower than some values reported from other DFT-based calculations (e.g., 9.67 eV).^[41] In contrast, the incorporation of aromatic moieties such as VN, DPE, and VBP led to pronounced changes in the electronic properties. The electron affinities became positive, increasing to 1.14 eV (VN), 0.69 eV (DPE), and 0.82 eV (VBP), indicating enhanced electron-accepting capabilities. Additionally, the ionization energies decreased relative to iPP, recorded as 5.88 eV (VN), 6.51 eV (DPE), and 6.38 eV (VBP). These trends reflect the influence of the electron-donating and π -resonance effects of the grafted aromatic groups on the molecular electronic states.

The changes in electronic potential levels resulting from the grafting of VN, DPE, and VBP onto the PP backbone were visualized using electrostatic potential (ESP) maps at an electron density iso surface of 0.002 a.u., as shown in **Figure 5**. For iPP (**Figure 5a**), the eleESP scale is confined within $\pm 6.83 \times 10^{-3}$ Hartree, indicating an almost electrostatically neutral distribution with no significant charge localization. In contrast, the PPs grafted with VN, DPE, and VBP exhibit a markedly expanded potential range of $\approx \pm 2.4$ to 2.8×10^{-2} Hartree. This quantitative increase in potential range supports the conclusion that the introduction of π -conjugated rings significantly enhances molecular electrostatic anisotropy and induces localized electric fields within the system.

In all grafted structures, distinct negative (red) and positive (blue) ESP regions are formed around the π -conjugated rings; however, their spatial distributions exhibit notable differences. In the case of VN-g-PP (**Figure 5b**), localized negative charge accumulation appears around the π -conjugated ring, but due to the presence of only a single conjugated plane, the extent of this negative region is relatively limited, resulting in the simplest characteristics in terms of electronic structural stability and polarizability. In contrast, although structurally similar at first glance, DPE-g-PP (**Figure 5c**) shows the formation of a continuous negative charge pocket as two phenyl rings are arranged in parallel through a vinyl linkage. This suggests a strongly delocalized electron cloud between the rings, indicating the potential for a concentrated donor region under an applied external electric field. On the other hand, in VBP-g-PP (**Figure 5d**), rotation around the single bond connecting the biphenyl rings leads to a partial twist

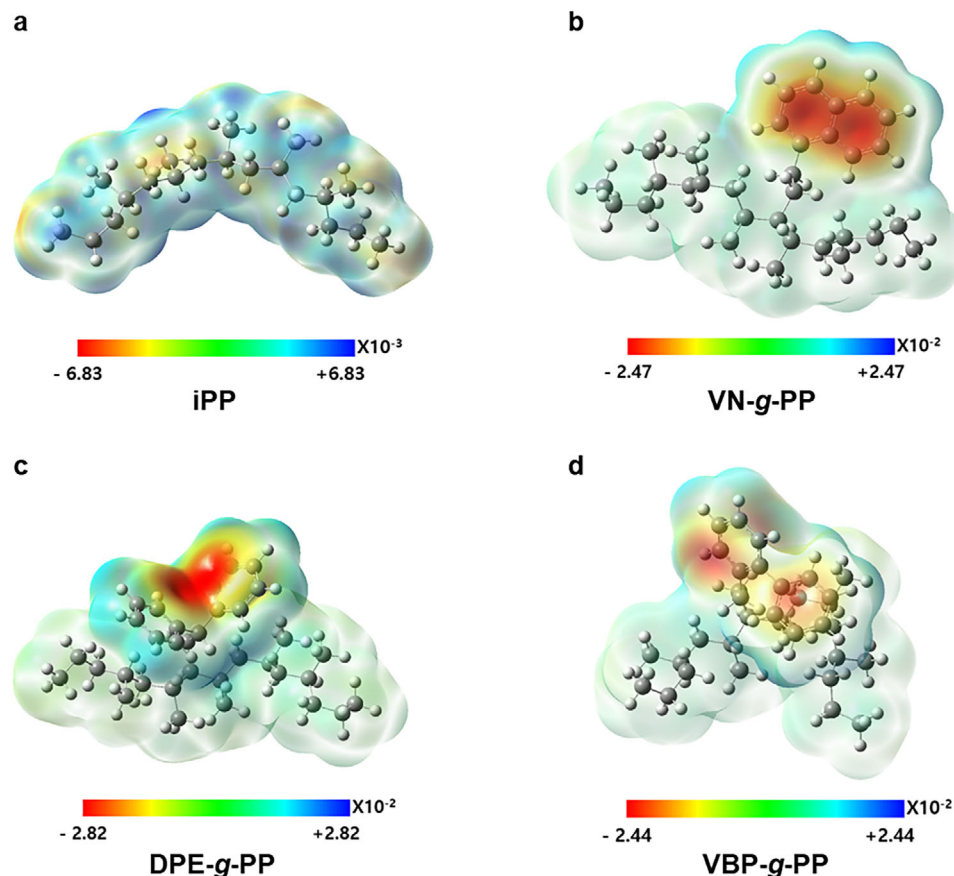


Figure 5. ESP of a) iPP, b) VN-g-PP, c) DPE-g-PP, and d) VBP-g-PP, respectively. [Hartree/electron].

in the π -conjugation plane. As a result, the negative charge pockets are segmented between ring 1 and ring 2. This torsion disrupts coplanarity, forming two distinct localized negative regions on each ring, which hinders the continuity of π -conjugation, reduces electron delocalization, and may negatively impact charge carrier mobility and internal electric field uniformity.

As shown in **Figure 6**, the partial density of states (PDOS) analysis of iPP and the VS grafted PPs reveals that distinct trap energy levels are induced depending on the molecular configuration. For ideal iPP (**Figure 6a**), no significant trap states appear within the bandgap, indicating limited intrinsic charge-trapping capability. In contrast, VN-g-PP (**Figure 6b**) exhibits a clearly defined deep trap level, isolated from the main bands, appearing as a sharp peak in the PDOS spectrum. This localized state is attributed to the π -conjugated naphthalene ring grafted onto the main chain, which introduces a strong electronic polarization.^[42] Notably, this deep trap formation aligns well with the TSDC results, where the trap depth increased to 1.21 eV upon the addition of 0.1 wt.% VN, confirming the effectiveness of the π - π interaction in enhancing charge localization through both theoretical and experimental approaches. For DPE-g-PP (**Figure 6c**), a multiple-trap structure is observed, comprising a distinct deep trap (≈ 0.7 eV) and several additional shallow traps (≈ 0.4 eV) near the band edges. These features arise from the cross-conjugated nature of the graft, which

creates a spatial separation between localized and partially conjugated orbitals. The bulky diphenylmethyl group introduces substantial steric hindrance, reducing interchain interaction and enabling charge carriers to remain trapped in individual localized sites for extended periods.^[43] In the case of VBP (**Figure 6d**), the PDOS spectrum shows several broad, shallow trap peaks located near the band edges, suggesting partially delocalized electronic states overlapping with the main bands. These shallow traps are likely induced by the torsional distortion between the biphenyl rings, which interrupts π -conjugation and creates localized perturbations near the band edges.^[44] This structural feature corresponds well with the dual-peak behavior observed in the TSDC data (1.15 and 1.23 eV at 0.1 wt.%) and the shallower trap depth (1.14 eV) found at higher VBP content, reinforcing the notion that increased grafting levels lead to poorer dispersion and reduced trap effectiveness due to structural disorder.

While the DFT calculations were performed under idealized gas-phase conditions, which may limit direct reflection of condensed-phase interactions in practical systems, the key trends in trap formation and energetic localization remain consistent with experimental TSDC results. These similarities highlight the complementary nature of theoretical and experimental approaches in elucidating charge trapping mechanisms in polymer dielectrics.

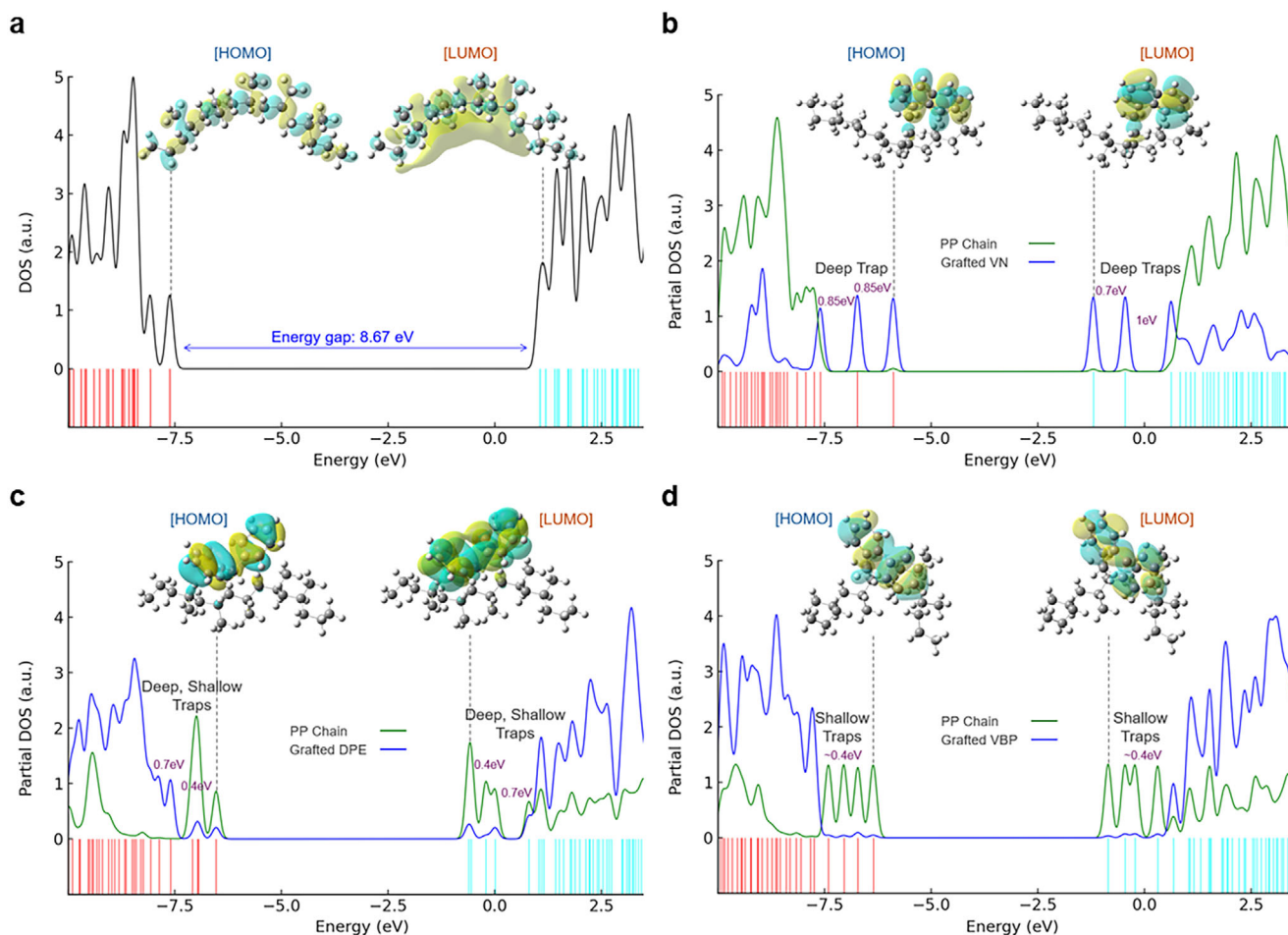


Figure 6. Energy diagram of a) iPP, b) VN-g-PP, c) DPE-g-PP, and d) VBP-g-PP, respectively.

2.6. Charge Transport Simulation: Time-Resolved Space Charge Evolution

In this study, the space charge transport characteristics of the grafted molecule were quantitatively evaluated by incorporating ESP distributions extracted from DFT calculations. In particular, the ESP distribution reflects both the depth and spatial distribution of local electric potential wells intrinsic to each molecule, which serve as key determinants of space charge trap energy and trapping probability. These DFT-based ESPs were integrated into the FEM simulation framework as boundary conditions, enabling a multiscale analysis of how localized trapping tendencies influence space charge transport and accumulation behavior within the iPP matrix (Figures S3 and S4, Supporting Information). A hole was injected from the anode, and an electron was injected from the cathode following the Schottky injection. This multiscale approach enables the translation of molecular-level trapping behavior into macroscopic space charge dynamics (detailed in Note S1, Supporting Information). **Figure 7** visualizes the time-dependent 3D spatial distribution of injected holes for VN, DPE, and VBP-grafted PP samples, at time points: 0.1, 1, 5, and 20 ms. For visual clarity, top-view variants at the same orientations, along with the

neat iPP control, are provided together (Figure S5, Supporting Information).

The pristine iPP exhibits a narrow ESP scale of $\pm 6.83 \times 10^{-3}$ Hartree (Figure 5a), providing only weak, shallow potential wells compared with the grafted cases. Consequently, injected holes are not strongly retained on the molecular surface and transport rapidly through the polymer matrix toward the opposite electrode. Under the same bipolar charge transport (BCT)/FEM settings, the effective conductivity of iPP is 110.56 pS m^{-1} , and the peak electric field reaches $18.879 \text{ MV mm}^{-1}$. This behavior contrasts with the grafted-molecule cases in Figure 7, where enlarged ESP ranges promote stronger localization and moderated field build-up.

In the VN-g-PP, injected holes are distinctly concentrated near the π -naphthalene ring starting as early as 0.1 ms, and this distribution remains stable throughout the simulation up to 20 ms. As shown in the ESP distribution of Figure 5b, the naphthalene ring exhibits a relatively uniform negative potential (down to -2.47×10^{-2} a.u.), indicating a stable trapping site for holes. Although the trap energy is not particularly high, the π -electron cloud inside the naphthalene ring guides the holes into this region of uniform electron density, which is consistently reproduced in the FEM simulation under applied fields. As a result,

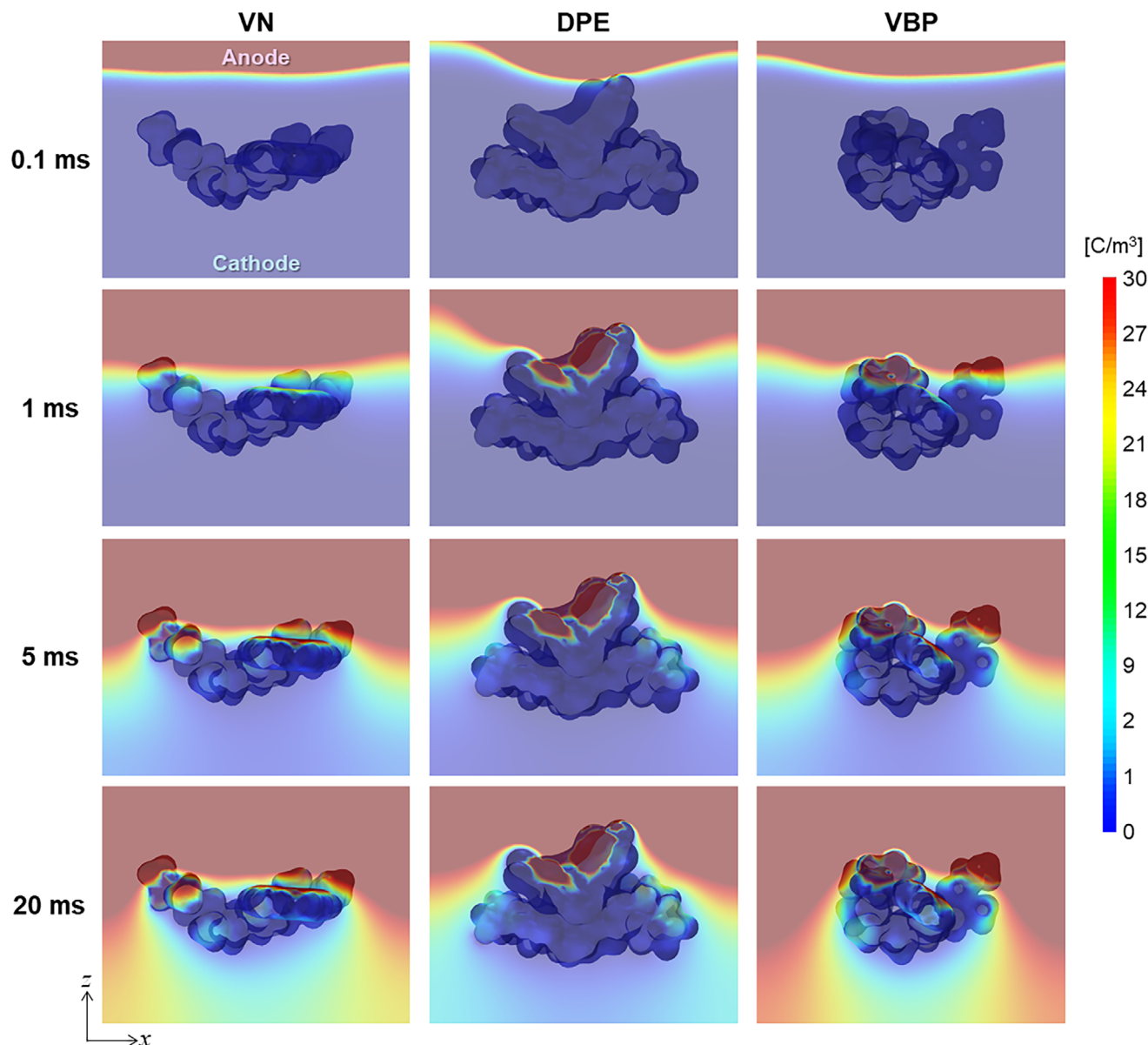


Figure 7. Schematic of space charge transport, specifically holes in grafted PP with iPP matrix. The cross-sectional hole density, C m^{-3} with VN, DPE, and VBP, based on Schottky injection process with 0.1, 1, 5, and 20 ms. The applied electric field is 350 MV m^{-1} and the Schottky barrier is fixed at 1.2 eV. The mobility in the polymer matrix was set at $1.0 \times 10^{-11} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for electron and hole, respectively. The semi-transparent blue molecular envelope indicates the 0.002 a.u. iso surface used to map DFT-derived ESP (Figure 5) into the continuum model. Top-view variants and the neat iPP analysis results are provided in Figure S5 (Supporting Information).

the holes are not confined near the electrode but instead stabilized within the grafted molecule and chain region, partially suppressing charge injection. The simulated conductivity was 76.1 pS m^{-1} , and the peak electric field reached 17.1 MV mm^{-1} , indicating a moderate level of insulation performance with improved hole confinement due to the central trap in the naphthalene group.

The DPE-g-PP displays a distinct space charge behavior characterized by relatively limited penetration of injected holes into the PP matrix compared to other samples. This behavior is strongly influenced by the parallel orientation of the DPE molecule con-

cerning the hole injection direction, thereby enabling direct access to the central conjugated π -system. As supported by the DFT-derived ESP distribution (Figure 5c), the molecule exhibits a localized negative ESP basin along the aromatic ring, which acts as an energetically favorable site for hole trapping. This alignment promotes efficient charge accommodation along the delocalized π -core while simultaneously restricting further transport into the bulk. Over time, the hole density remains concentrated around the DPE structure with relatively weak diffusion into deeper regions of the polymer matrix, indicating a strong localization effect. Compared to VN and VBP samples, which

exhibit a broader spread of charge into the matrix, DPE shows the most spatially confined charge profile, with the lowest field intensity (16.2 MV mm^{-1}) and conductivity (63.2 pS m^{-1}). This behavior suggests that the DPE's molecular topology and electrostatic environment jointly suppress excessive space charge accumulation and long-range charge transport, contributing to the material's excellent dielectric breakdown resistance.

The VBP-g-PP initially shows significant hole accumulation at the centre of the grafted molecule. According to the ESP distribution in Figure 5d, the VBP ring core exhibits a homogeneous negative potential ($\approx -2.44 \times 10^{-2} \text{ a.u.}$), forming a potential pocket favorable for hole capture. This phenomenon is supported by the FEM simulation, which shows strong hole concentration in the molecular core at 0.1 ms. However, due to insufficient trap energy, holes gradually diffuse toward the molecular periphery and bulk direction over time, eventually reaching the opposite electrode side by 20 ms. As a result, electric field concentration occurs at both ends of the material, leading to the strong electric field (18.4 MV mm^{-1}) and comparably high conductivity (85.8 pS m^{-1}). Although the molecule initially supports centralized trapping, it results in extensive diffusion, making this structure the most vulnerable in terms of insulation reliability.

To extend the molecular-scale simulations to bulk space-charge behavior, we performed the space charge measurement with time in iPP and the 0.5 wt.% VN/DPE/VBP-g-PP samples based on pulsed electroacoustic method. Although measured bulk space charge dynamics do not mirror the simulated space-charge dynamics one-to-one, they are qualitatively informative. In the measurement results, the positive-peak (hole) order is $\text{VN} > \text{DPE} \approx \text{iPP} > \text{VBP}$, whereas the negative-peak (electron) order is $\text{VN} \gtrsim \text{VBP} \gtrsim \text{DPE} \approx \text{iPP}$ (Figure S6, Supporting Information). Consistent with Figure 7, deep-trap-mediated hole capture is strongest in VN—and appreciable in DPE—so a larger fraction of injected holes remains localized near the grafted π -cores; this is well aligned with the higher positive peaks observed for VN (and, relative to VBP, for DPE) in the space-charge measurement results. By contrast, VBP exhibits the lowest positive peak and post-capture spreading of carriers: as indicated by the newly introduced shallow states in the DFT analysis (Figure 6c,d), holes that are initially injected subsequently propagate through the matrix, producing a smaller positive peak through the opposite electrode. This is also consistent with the transport metrics from the simulations in Figure 7—among the grafted systems, VBP shows the highest conductivity ($\approx 85.8 \text{ pS m}^{-1}$). Together with the electrode work-function asymmetry (Cu anode $\approx 4.70 \text{ eV}$; Al cathode $\approx 4.28 \text{ eV}$) and grafting-induced frontier-level shifts (lower ionization energy and positive electron affinity for VN/DPE/VBP vs iPP; Table 4), the injection barrier is lowest for VN and highest for iPP, which explains both the positive/negative peaks asymmetry at 0.5 wt.% samples.

3. Conclusion

This work demonstrates a practical, one-step melt grafting strategy for covalently incorporating geometry-tailored aromatic VSs into iPP. VN, DPE, and VBP were systematically selected and successfully grafted based on their melt-processable vinyl functionality, aromatic π -conjugation, and thermal stability. Electrical

characterization confirmed that VS grafting effectively enhanced the insulation performance of iPP.

TSDC and quantum chemical simulations revealed that VS geometry critically governs trap formation and charge localization, where VN and DPE facilitated deeper and more uniform traps, while VBP showed limited benefit due to less favorable geometry. At higher VS contents (VN and DPE), morphological analyses revealed local aggregation and phase separation, indicating that such structural inhomogeneity can also influence the resulting electrical behavior beyond trap characteristics.

These results establish a clear link between VS molecular design, charge-trapping behaviors, and macroscopic insulation performance, providing molecular-level insight into how stabilizer geometry influences insulation reliability. Overall, this study validates melt grafting as a scalable, solvent-free route for functionalizing iPP and highlights the potential for rationally designing melt-graftable VSs optimized for charge trapping. Nevertheless, further efforts are still required to realize large-scale implementation, particularly in optimizing processing parameters, materials composition, and initiator control to ensure uniform dispersion and stable grafting performance. In this context, this approach provides a foundation for developing recyclable thermoplastic insulation tailored for next-generation HVDC systems.

4. Experimental Section

Sample Preparation: iPP (HP450J, Polymirae, Korea) was simultaneously melt-mixed with 0.1 phr of DCP (329 541, Sigma-Aldrich, USA) as the initiator and a specified ratio of voltage stabilizer—VN (V2909), VBP (V1805) or DPE (D206806) (all from Sigma-Aldrich, USA)—using an internal mixer (W50EHT+ Plastograph EC, Brabender GmbH, Germany) at 180°C and 60 rpm for 15 min. The resulting mixture was then placed into a stainless-steel mold and hot-pressed at 180°C and 10 MPa for 5 min, followed by rapid cooling to 10°C for 2 min.

Leakage Current Measurement: Leakage current measurements were conducted using a standard three-electrode system enclosed in an insulated heating oven (WiseVen, DAIHAN Scientific, Korea). The system was connected to an electrometer (6517A, Keithley, USA) and a high-voltage power supply (2290E-5, Keithley, USA). All tests were performed at 25°C . For each measurement, samples were subjected to constant applied voltages of 0.5, 1.0, and 1.5 kV each to evaluate field-dependent conduction behavior. The leakage current was recorded for 1800 s under each condition, and the raw data were first processed using a moving median filter followed by a moving average filter to remove high-frequency noise.^[45] The steady-state current (I) was then determined by averaging the filtered data over the final 300 s after stabilization. The volume resistivity (ρ) was subsequently calculated from the steady-state leakage current using the relation:

$$\rho = \frac{V \times A}{I \times t} \quad (1)$$

where V is the applied voltage, A is the electrode area, and t is the specimen thickness.

DC Breakdown Test: DC BD tests were performed in accordance with the IEC 60243-1 standard using a dielectric strength tester (Sky Innotek Co., Ltd, Korea). Measurements were conducted at 25°C . The test setup utilized a pair of stainless-steel ball electrodes, each with a diameter of 20 mm. To prevent surface flashover in air, the sample and electrodes were immersed in silicone oil during testing. For each sample type, ten independent BD tests were conducted to ensure statistical reliability. The BD voltage (BDV) was recorded for each test, and the BDS was calculated by dividing the BDV by the sample thickness.

TSDC Measurement: TSDC measurements were conducted to analyze the trap level distribution in the samples using a TSDC measurement system (Toyo Corporation, Japan) integrated with a high-precision electrometer (B2987A, Keysight, USA). Film specimens with a thickness of $100 \pm 15 \mu\text{m}$ were cut to dimensions of $20 \times 20 \text{ mm}$, and gold electrodes (10 mm in diameter) were deposited on both sides via ion sputtering to ensure uniform electrical contact. The samples were polarized under an electric field of -1 kV mm^{-1} at 110°C for 30 min, followed by rapid cooling to -60°C using liquid nitrogen to freeze the trapped charges. The electric field was then removed, and the samples were short-circuited for 1 min. Subsequently, the temperature was increased to 160°C at a constant heating rate of 3°C min^{-1} while the depolarization current was recorded. Trap level distributions were extracted from the TSDC data using a custom MATLAB script.^[46]

PEA Measurement: Space charge measurements were conducted using a flat-sample-type pulsed electroacoustic (PEA) system (Techimp, Italy). Sheet samples with a thickness of $500 \mu\text{m}$ were tested under a uniform DC field of 20 kV mm^{-1} , corresponding to an applied voltage of 10 kV. A built-in pulse generator provided 10 ns-wide, 200 V measurement pulses, which were identical for all samples. The protocol consisted of 1 h charging under DC bias followed by 1 h discharging under short-circuit conditions to evaluate accumulated and residual charges. The detected raw signal was corrected for attenuation and dispersion using an in-house deconvolution algorithm to obtain the actual space charge density distribution.

General Characterization: The fracture surfaces were analyzed using field emission scanning electron microscopy (FE-SEM, S-4800, Hitachi, Japan). UV-vis spectra were measured using UV-vis spectrophotometer (V-770, JASCO, Japan) in the range from 200 to 800 nm.

Quantum Chemical Simulation: Quantum chemical calculations based on DFT were conducted to investigate the electronic structures of iPP and its derivatives: VN-g-PP, DPE-g-PP, and VBP-g-PP. All computations were carried out using the Gaussian 16 commercial software package. This analysis focused on comparing the electronic properties, such as electron distribution, and the highest occupied molecular orbital (HOMO)–lowest unoccupied molecular orbital (LUMO) energy gap, with particular emphasis on elucidating charge transport behavior under high electric field conditions. Geometry optimization was performed for the molecular structures illustrated in Figure S7 (Supporting Information).

The calculations employed the B3LYP functional with the 6–311G (d,p) basis set, and Grimme's D3BJ empirical dispersion correction was applied to account for van der Waals interactions.^[47,48] To enhance the accuracy of the calculations, the convergence criteria were set to a tight level: the thresholds for energy change (ΔE), gradient, and displacement were restricted to 10^{-6} Hartree, 3×10^{-4} Hartree/Bohr, and 1.2×10^{-3} Bohr, respectively. For stable convergence of the electronic structure calculations, the self-consistent field procedure was performed with an energy change threshold of less than 10^{-8} Hartree. Electronic structure analyses were subsequently conducted on the optimized geometries without further structural modification.

Charge Transport Simulation: The internal Dirichlet boundary conditions in a dielectric material were defined based on the ESP distribution calculated at the single molecular level, and were linked to a FEM-based BCT model to numerically analyze charge dynamics within the medium.^[49] The electrostatic potential distribution on the 0.002 a.u. The electron density iso surface was calculated using the Multiwfn program,^[50,51] based on wavefunction outputs from Gaussian 16. Local regions with high electron affinity were regarded as sites for the formation of electron trap levels, and the trap density and energy were assigned a spatially non-uniform distribution accordingly. This ESP-based boundary information was incorporated into the user-defined partial differential equation module of COMSOL Multiphysics 6.2. Within the BCT equations, the mobility, trap capture and release coefficients, and recombination terms were implemented with spatial dependence. (The governing equations and coefficients used are described in the Supporting Information.) The continuity equations for electrons and holes, together with the Poisson equation, were formulated as a coupled system and solved to obtain the time-dependent charge density and electric-field distributions. Through this approach, it was enabled

a qualitative evaluation of how the molecular structural anisotropy of the actual material influences macroscopic charge behavior and the formation of field distortion.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

high-voltage direct current, insulation, melt grafting, polypropylene, voltage stabilizers

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